Unit: Unit 0: Foundations of Algebra
Topic: Equations

Name: ________________
Date: ___________

Quiz (Habanero)

Q1. Solve for x: $x + 5 = 11$

(A) $x = 6$
(B) $x = 7$
(C) $x = 8$
(D) $x = 9$
(E) $x = 10$

Q2. A glacier is losing 3 meters of ice per year. If it currently has 45 meters of ice, how many years will it take for the glacier to lose 15 meters of ice?

(A) 5 years
(B) 3 years
(C) 2 years
(D) 1 year
(E) 4 years

Q3. The rainfall in a desert region is increasing by 2 inches per month. If the current rainfall is 10 inches, how many months will it take for the rainfall to reach 16 inches?

(A) 5 months
(B) 3 months
(C) 2 months
(D) 6 months
(E) 4 months

Q4. Solve for y: $2y - 4 = 12$

(A) $y = 6$
(B) $y = 7$
(C) $y = 8$
(D) $y = 9$
(E) $y = 10$

Q5. The temperature in a greenhouse is rising by 1.5 degrees Celsius per hour. If the current temperature is 20 degrees Celsius, how many hours will it take for the temperature to reach 25 degrees Celsius?

(A) 5 hours
(B) 3 hours
(C) 2 hours
(D) 1.67 hours
(E) 4 hours

Q6. Solve for z: $z/2 + 3 = 7$

(A) $z = 6$
(B) $z = 7$
(C) $z = 8$
(D) $z = 9$
(E) $z = 10$

Q7. A lake is being filled at a rate of 0.5 meters per day. If the lake currently has 2 meters of water and needs 4 meters to be considered full, how many days will it take to fill the lake?

(A) 2 days
(B) 3 days
(C) 4 days
(D) 6 days
(E) 8 days

Q8. Solve for x: $x - 2 = 9$

(A) $x = 7$
(B) $x = 10$
(C) $x = 11$
(D) $x = 12$
(E) $x = 13$

Open Ended Questions (FRQ)

Q9. Solve for x: $2x + 5 = 11$. Show all work and check your solution for accuracy.

Q10. A river is flowing at a rate of 1.2 meters per second. If the river currently has 50 meters of water and is losing 0.5 meters per hour due to evaporation, how many hours will it take for the river to lose 10 meters of water? Show all work and check your solution for accuracy.

Chef's Tips

- Tip 1: Encourage students to use algebraic methods to solve equations.
- Tip 2: Emphasize the importance of checking solutions for accuracy.
- Tip 3: Consider providing additional support for students struggling with multi-step linear equations.